Econ 101A
Section 11: Solutions

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Last time we stopped at the cost function. Today we finish the firm. We pick output, read the supply curve off the cost curves, and then solve the whole thing in one step to check that we get the same answer. One theme runs through everything: returns to scale decide whether the firm's problem has a nice interior answer. Under perfect competition, only decreasing returns give a well-behaved supply curve.

1 Cost Minimization, Step 2: Choosing Output

Given the cost function $c(w, r, y)$ and an output price p the firm takes as given, the firm solves

$$\max_{y \geq 0} \; py - c(w, r, y).$$

First-order condition. $p - c'_y(w, r, y) = 0$, that is, **price equals marginal cost**.

Second-order condition. $-c''_{yy}(w, r, y^*) < 0$, that is, marginal cost must be *increasing* at y^*. If marginal cost is falling, the first-order condition finds a profit minimum.

Supply slopes up. The supply function $y^*(w, r, p)$ is defined by $F(y, p) \equiv p - c'_y(y) = 0$. By the implicit function theorem, with $F_p = 1$ and $F_y = -c''_{yy}$,

$$\frac{\partial y^*}{\partial p} = -\frac{F_p}{F_y} = \frac{1}{c''_{yy}} > 0 \quad \text{whenever the SOC holds.}$$

Returns to scale shape costs. For $f = AK^\alpha L^\beta$, Lecture 8 shows that

$$c(w, r, y) = B\Big(\frac{y}{A}\Big)^{\frac{1}{\alpha+\beta}}, \qquad B \equiv w\Big(\frac{w\alpha}{r\beta}\Big)^{-\frac{\alpha}{\alpha+\beta}} + r\Big(\frac{w\alpha}{r\beta}\Big)^{\frac{\beta}{\alpha+\beta}}.$$

The exponent $1/(\alpha + \beta)$ does all the work:

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- DRS ($\alpha + \beta < 1$): exponent above 1, cost is convex, MC rises. Interior optimum, supply well defined.
- CRS ($\alpha + \beta = 1$): cost is linear in y, MC is flat at B/A. Produce nothing if $p < B/A$, anything if $p = B/A$, "infinitely much" if $p > B/A$.
- IRS ($\alpha + \beta > 1$): cost is concave, MC falls. The FOC gives a minimum; profit grows without bound in y.

2 Cost Curves and the Supply Function

Marginal cost $\text{MC} = \partial c/\partial y$ tells you how much to produce: $p = \text{MC}$.

Average cost $\text{AC} = c/y$ tells you whether to produce at all. Dividing profit by $y > 0$,

$$\pi = py - c > 0 \iff p - \frac{c}{y} > 0 \iff \text{AC} < p.$$

MC cuts AC at its minimum. Differentiate AC with the quotient rule:

$$\frac{d}{dy}\Big(\frac{c(y)}{y}\Big) = \frac{c'(y)\,y - c(y)}{y^2} = \frac{1}{y}\Big(c'(y) - \frac{c(y)}{y}\Big) = \frac{\text{MC} - \text{AC}}{y}.$$

AC falls when MC is below it and rises when MC is above it. So the two curves cross where AC is flat, at its minimum.

The supply curve is the part of the MC curve that lies above AC. Below minimum AC, the firm shuts down and supplies zero.

3 One-Step Profit Maximization

Instead of two steps, choose inputs directly:

$$\max_{L,K}\ pf(L,K) - wL - rK.$$

First-order conditions. $pf'_L = w$ and $pf'_K = r$. Each input is hired until the *value of its marginal product* equals its price. Dividing one by the other gives back $f'_L/f'_K = w/r$: the one-step firm also minimizes cost.

Second-order conditions. $pf''_{LL} < 0$ and

$$|H| = \begin{vmatrix} pf''_{LL} & pf''_{LK} \\ pf''_{LK} & pf''_{KK} \end{vmatrix} = p^2\Big[f''_{LL}f''_{KK} - \big(f''_{LK}\big)^2\Big] > 0.$$

Diminishing marginal products are not enough. The cross effect f''_{LK} must not be too large.

Comparative statics. Differentiate the two FOCs with respect to w and to r, and solve with Cramer's rule:

$$\frac{\partial L^*}{\partial w} = \frac{p f''_{KK}}{|H|} < 0, \qquad \frac{\partial L^*}{\partial r} = \frac{-p f''_{LK}}{|H|}.$$

Labor demand always falls with its own price. The effect of the price of capital depends on the sign of f''_{LK}: if capital makes workers more productive ($f''_{LK} > 0$), a higher r lowers labor demand too.

4 Exercises

Exercise 1 (a well-behaved firm) Let $y = K^{1/4}L^{1/4}$, input prices $w, r > 0$, output price $p > 0$.

(a) What are the returns to scale? Find the conditional factor demands and the cost function.

(b) Find MC and AC. Which is larger?

(c) Solve for the supply function $y^*(w, r, p)$. Check the SOC and verify $\partial y^*/\partial p = 1/c''_{yy}$.

(d) Let $w = r = 1$ and $p = 8$. Find y^*, L^*, K^* and profit.

Solution. **(a)** $f(tL, tK) = (tK)^{1/4}(tL)^{1/4} = t^{1/2}f$. Since $1/2 < 1$, DRS.

Tangency. With $\alpha = \beta = \frac{1}{4}$, Section 10 gives MRTS $= \frac{\beta}{\alpha}\frac{K}{L} = \frac{K}{L}$. Set it equal to w/r: $K = \frac{w}{r}L$. Put this into the constraint:

$$\Big(\frac{w}{r}L\Big)^{1/4}L^{1/4} = y \implies \Big(\frac{w}{r}\Big)^{1/4}L^{1/2} = y \implies L^{1/2} = y\Big(\frac{r}{w}\Big)^{1/4} \implies L^* = y^2\Big(\frac{r}{w}\Big)^{1/2}.$$

Then $K^* = \frac{w}{r}L^* = y^2\big(\frac{w}{r}\big)\big(\frac{w}{r}\big)^{-1/2} = y^2\big(\frac{w}{r}\big)^{1/2}$. Cost:

$$c = wL^* + rK^* = w\, y^2 r^{1/2} w^{-1/2} + r\, y^2 w^{1/2} r^{-1/2} = y^2\sqrt{wr} + y^2\sqrt{wr} = 2y^2\sqrt{wr}.$$

(Check with the lecture formula: $1/(\alpha+\beta) = 2$, so cost is proportional to y^2.)

(b)

$$\text{MC} = \frac{\partial c}{\partial y} = 4y\sqrt{wr}, \qquad \text{AC} = \frac{c}{y} = \frac{2y^2\sqrt{wr}}{y} = 2y\sqrt{wr}.$$

Both rise with y, and MC $= 2\,$AC $>$ AC for all $y > 0$. Since MC is always above AC, AC is always rising, which matches the formula.

(c) The firm maximizes $py - 2y^2\sqrt{wr}$. FOC:

$$p - 4y\sqrt{wr} = 0 \implies y^* = \frac{p}{4\sqrt{wr}}.$$

SOC: $\frac{d^2}{dy^2}\big(py - 2y^2\sqrt{wr}\big) = -4\sqrt{wr} < 0$. It is a maximum. Then

$$\frac{\partial y^*}{\partial p} = \frac{1}{4\sqrt{wr}}, \qquad c''_{yy} = \frac{\partial}{\partial y}\big(4y\sqrt{wr}\big) = 4\sqrt{wr},$$

so indeed $\partial y^*/\partial p = 1/c''_{yy}$. Since $p = \text{MC} > \text{AC}$ at y^*, profit is positive: the firm always produces.

(d) With $w = r = 1$, $\sqrt{wr} = 1$.

$$y^* = \frac{8}{4} = 2, \qquad L^* = 2^2 \cdot 1 = 4, \qquad K^* = 2^2 \cdot 1 = 4.$$

Check output: $4^{1/4} \cdot 4^{1/4} = 4^{1/2} = 2$. Cost $= wL^* + rK^* = 4 + 4 = 8$, which matches $2y^2\sqrt{wr} = 2 \cdot 4 = 8$. Revenue $= py^* = 16$. Profit $= 16 - 8 = 8$.

More generally, $\pi^* = p\,\frac{p}{4s} - 2\big(\frac{p}{4s}\big)^2 s$ with $s = \sqrt{wr}$. Square the fraction, $2 \cdot \frac{p^2}{16s^2} \cdot s = \frac{p^2}{8s}$, so $\pi^* = \frac{p^2}{4s} - \frac{p^2}{8s} = \frac{p^2}{8s}$. At $p = 8$, $s = 1$: $64/8 = 8$. And $\partial\pi^*/\partial p = \frac{2p}{8s} = \frac{p}{4s} = y^*$, the envelope theorem (Hotelling's lemma).

Exercise 2 (what goes wrong with CRS and IRS)

(a) Let $y = K^{1/2}L^{1/2}$. In Section 10 we found $c = 2y\sqrt{wr}$. Find MC and AC, and solve the output choice for every p.

(b) Let $y = KL$. Find the conditional factor demands and the cost function. Find the y that solves the FOC. Is it a maximum? What is profit there? What does the firm do?

Solution. **(a)** MC $= \partial c/\partial y = 2\sqrt{wr}$ and AC $= c/y = 2\sqrt{wr}$. Both are flat and equal. Profit is

$$\pi(y) = py - 2y\sqrt{wr} = \big(p - 2\sqrt{wr}\big)\,y,$$

a straight line in y. The FOC $p - 2\sqrt{wr} = 0$ does not contain y, and $c''_{yy} = 0$. Three cases:

- $p > 2\sqrt{wr}$: each unit earns a positive margin, so profit grows forever. $y^* \to \infty$.
- $p = 2\sqrt{wr}$: profit is zero for every y. Any $y^* \in [0, \infty)$.
- $p < 2\sqrt{wr}$: each unit loses money. $y^* = 0$.

This matches the lecture's B/A: with $\alpha = \beta = \frac{1}{2}$ and $A = 1$, $B = w(w/r)^{-1/2} + r(w/r)^{1/2} = \sqrt{wr} + \sqrt{wr} = 2\sqrt{wr}$. Under CRS there is no supply *function*, only a horizontal supply curve at $p = B/A$.

(b) Marginal products: $f'_L = K$, $f'_K = L$. Tangency: $\frac{K}{L} = \frac{w}{r}$, so $K = \frac{w}{r}L$. Constraint:

$$\frac{w}{r}L \cdot L = y \implies L^2 = \frac{ry}{w} \implies L^* = \Big(\frac{ry}{w}\Big)^{1/2}, \qquad K^* = \frac{w}{r}\Big(\frac{ry}{w}\Big)^{1/2} = \Big(\frac{wy}{r}\Big)^{1/2}.$$

Cost:

$$c = w\Big(\frac{ry}{w}\Big)^{1/2} + r\Big(\frac{wy}{r}\Big)^{1/2} = \sqrt{wry} + \sqrt{wry} = 2\sqrt{wr}\,y^{1/2}.$$

Then MC $= 2\sqrt{wr} \cdot \frac{1}{2}y^{-1/2} = \sqrt{wr}\,y^{-1/2}$, which falls with y. FOC:

$$p = \sqrt{wr}\,y^{-1/2} \implies y^{1/2} = \frac{\sqrt{wr}}{p} \implies y^{\text{foc}} = \frac{wr}{p^2}.$$

SOC: $c''_{yy} = -\frac{1}{2}\sqrt{wr}\,y^{-3/2} < 0$, so $-c''_{yy} > 0$. The FOC point is a profit *minimum*. Profit there:

$$p\,\frac{wr}{p^2} - 2\sqrt{wr}\,\Big(\frac{wr}{p^2}\Big)^{1/2} = \frac{wr}{p} - 2\sqrt{wr}\,\frac{\sqrt{wr}}{p} = \frac{wr}{p} - \frac{2wr}{p} = -\frac{wr}{p} < 0.$$

As y grows, $\pi = py - 2\sqrt{wr}\, y^{1/2}$ is dominated by the linear term, so $\pi \to \infty$. The firm wants $y^* \to \infty$. Example with $w = r = p = 1$: at $y = 1$, $\pi = 1 - 2 = -1$; at $y = 16$, $\pi = 16 - 2 \cdot 4 = 8$. With falling average costs, one big firm beats many small ones. Perfect competition cannot handle that, which is why increasing returns usually come with market power.

Exercise 3 (cost curves with one input and a setup cost) One input, labor, with wage w. Technology $y = L^\alpha$, $\alpha > 0$.

(a) Find $c(w, y)$, MC and AC. Show that $\text{MC} = \text{AC}/\alpha$.

(b) Let $\alpha = \frac{1}{2}$ and $w = 2$. Find the supply function.

(c) Describe MC, AC and supply when $\alpha = 1$ and when $\alpha > 1$.

(d) Keep $\alpha = \frac{1}{2}$, $w = 2$, but producing any positive output requires a setup cost $F = 8$, which the firm avoids if it produces nothing. Find MC, AC, the minimum of AC, and the supply function. Sketch the curves.

Solution. **(a)** To produce y you need $L = y^{1/\alpha}$, so $c(w, y) = wy^{1/\alpha}$. Then

$$\text{MC} = \frac{w}{\alpha}\, y^{\frac{1}{\alpha}-1} = \frac{w}{\alpha}\, y^{\frac{1-\alpha}{\alpha}}, \qquad \text{AC} = \frac{wy^{1/\alpha}}{y} = w\, y^{\frac{1}{\alpha}-1} = w\, y^{\frac{1-\alpha}{\alpha}},$$

so $\text{MC} = \text{AC}/\alpha$.

(b) With $\alpha = \frac{1}{2}$, $w = 2$: $c = 2y^2$, $\text{MC} = 4y$, $\text{AC} = 2y$. Setting $p = 4y$ gives

$$y^* = \frac{p}{4}.$$

SOC: $c'' = 4 > 0$. At y^*, $\text{AC} = 2 \cdot \frac{p}{4} = \frac{p}{2} < p$, so profit is positive. In general, for $\alpha < 1$, $p = \frac{w}{\alpha} y^{(1-\alpha)/\alpha}$ gives $y^{(1-\alpha)/\alpha} = \frac{\alpha p}{w}$ and $y^* = \left(\frac{\alpha p}{w}\right)^{\alpha/(1-\alpha)}$. With $\alpha = \frac{1}{2}$, $w = 2$: $\left(\frac{p}{4}\right)^1$, as above.

(c) $\alpha = 1$ (CRS): $c = wy$, $\text{MC} = \text{AC} = w$, both flat. Supply is horizontal at $p = w$: zero below, anything at $p = w$, unbounded above. Same as Exercise 2(a).

$\alpha > 1$ (IRS): the exponent $(1-\alpha)/\alpha$ is negative, so MC and AC both fall, and $\text{MC} = \text{AC}/\alpha < \text{AC}$. The FOC point is a profit minimum and the firm wants unbounded output. Same as Exercise 2(b).

(d) For $y > 0$, $c(y) = 8 + 2y^2$; and $c(0) = 0$. Then

$$\text{MC} = 4y, \qquad \text{AC} = \frac{8 + 2y^2}{y} = \frac{8}{y} + 2y.$$

The setup cost does not change MC, but it makes AC U-shaped. Minimize AC:

$$\frac{d\,\text{AC}}{dy} = -\frac{8}{y^2} + 2 = 0 \implies y^2 = 4 \implies y = 2, \qquad \text{AC}(2) = \frac{8}{2} + 2 \cdot 2 = 8.$$

Check: $\text{MC}(2) = 8 = \text{AC}(2)$. MC cuts AC at its minimum.

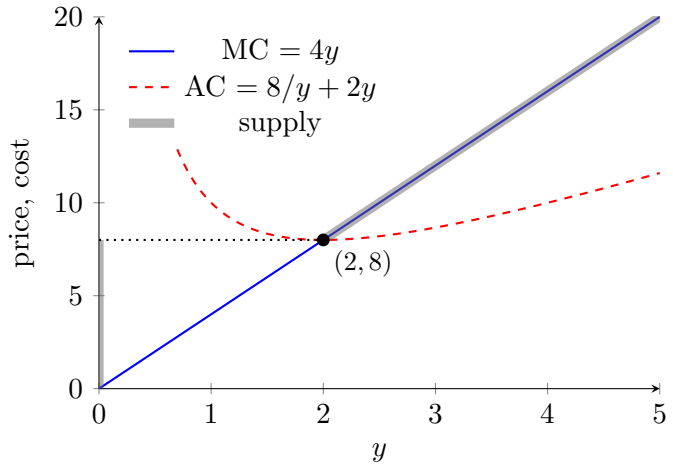
Supply. If the firm produces, it sets $p = 4y$, so $y = p/4$. Profit there is

$$\pi = p \cdot \frac{p}{4} - 8 - 2\left(\frac{p}{4}\right)^2 = \frac{p^2}{4} - \frac{2p^2}{16} - 8 = \frac{p^2}{4} - \frac{p^2}{8} - 8 = \frac{p^2}{8} - 8.$$

This is nonnegative iff $p^2 \geq 64$, iff $p \geq 8$, which is exactly minimum AC. So

$$y^*(p) = \begin{cases} p/4 & \text{if } p \geq 8, \\ 0 & \text{if } p < 8. \end{cases}$$

Supply jumps from 0 to 2 at $p = 8$: it is the MC curve above minimum AC.



(Prices go on the vertical axis. For $p < 8$ the firm sits at $y = 0$; for $p \geq 8$ it moves along MC.)

Exercise 4 (one step, same answer) Take the firm of Exercise 1, $y = K^{1/4}L^{1/4}$, and solve $\max_{L,K}\ pK^{1/4}L^{1/4} - wL - rK$ directly.

(a) Write the FOCs and solve for $L^*(w, r, p)$ and $K^*(w, r, p)$. Check against Exercise 1(d).

(b) For general $f = AK^\alpha L^\beta$, show that $f''_{LL}f''_{KK} - (f''_{LK})^2 = \alpha\beta(1-\alpha-\beta)A^2K^{2\alpha-2}L^{2\beta-2}$. When do the SOCs hold?

(c) Sign $\partial L^*/\partial w$ and $\partial L^*/\partial r$. Compare with the conditional demand $L^*(w, r, y) = y^2(r/w)^{1/2}$ from Exercise 1.

(d) At $p = 8$, $w = r = 1$, verify that $\left.\dfrac{\partial L^*}{\partial r}\right|_{\text{one-step}} = \left.\dfrac{\partial L^*}{\partial r}\right|_{y \text{ fixed}} + \left.\dfrac{\partial L^*}{\partial y}\right|_{\text{cond.}} \cdot \dfrac{\partial y^*}{\partial r}$. Interpret.

Solution. **(a)** FOCs:

$$\text{(L)}\ \ \frac{p}{4}K^{1/4}L^{-3/4} = w, \qquad \text{(K)}\ \ \frac{p}{4}K^{-3/4}L^{1/4} = r.$$

Divide (L) by (K). The $\frac{p}{4}$ cancels:

$$\frac{K^{1/4}L^{-3/4}}{K^{-3/4}L^{1/4}} = K^{1/4+3/4}L^{-3/4-1/4} = \frac{K}{L} = \frac{w}{r} \implies K = \frac{w}{r}L.$$

Put this into (L):

$$\frac{p}{4}\left(\frac{w}{r}\right)^{1/4}L^{1/4}L^{-3/4} = w \implies \frac{p}{4}\left(\frac{w}{r}\right)^{1/4}L^{-1/2} = w \implies L^{1/2} = \frac{p}{4}\,\frac{w^{1/4}r^{-1/4}}{w} = \frac{p}{4}\,w^{-3/4}r^{-1/4}.$$

Square both sides:

$$L^* = \frac{p^2}{16}\, w^{-3/2} r^{-1/2}, \qquad K^* = \frac{w}{r} L^* = \frac{p^2}{16}\, w^{-1/2} r^{-3/2}.$$

At $p = 8$, $w = r = 1$: $L^* = K^* = \frac{64}{16} = 4$, exactly as in Exercise 1(d). Output: $y = (K^* L^*)^{1/4} = \big(\frac{p^4}{256} w^{-2} r^{-2}\big)^{1/4} = \frac{p}{4\sqrt{wr}}$, the supply function from Exercise 1(c).

(b) The second derivatives are

$$f''_{LL} = \beta(\beta-1) A K^{\alpha} L^{\beta-2}, \qquad f''_{KK} = \alpha(\alpha-1) A K^{\alpha-2} L^{\beta}, \qquad f''_{LK} = \alpha\beta A K^{\alpha-1} L^{\beta-1}.$$

Multiply the first two. The powers add up to $K^{2\alpha-2} L^{2\beta-2}$:

$$f''_{LL} f''_{KK} = \alpha\beta(\alpha-1)(\beta-1) A^2 K^{2\alpha-2} L^{2\beta-2}.$$

Square the cross term: $(f''_{LK})^2 = \alpha^2\beta^2 A^2 K^{2\alpha-2} L^{2\beta-2}$. Subtract and factor out $\alpha\beta A^2 K^{2\alpha-2} L^{2\beta-2}$:

$$f''_{LL} f''_{KK} - (f''_{LK})^2 = \alpha\beta A^2 K^{2\alpha-2} L^{2\beta-2} \big[(\alpha-1)(\beta-1) - \alpha\beta\big].$$

Expand the bracket: $(\alpha-1)(\beta-1) - \alpha\beta = \alpha\beta - \alpha - \beta + 1 - \alpha\beta = 1 - \alpha - \beta$. This gives the formula. The SOCs need $f''_{LL} < 0$, so $\beta < 1$, and the determinant positive, so $\alpha + \beta < 1$. Since $\alpha + \beta < 1$ already forces $\beta < 1$, the SOCs hold exactly under DRS. Same message as the two-step approach. Here $\alpha = \beta = \frac{1}{4}$: $\alpha\beta(1 - \alpha - \beta) = \frac{1}{16} \cdot \frac{1}{2} = \frac{1}{32} > 0$.

(c) From $L^* = \frac{p^2}{16} w^{-3/2} r^{-1/2}$:

$$\frac{\partial L^*}{\partial w} = \frac{p^2}{16}\Big(-\frac{3}{2}\Big) w^{-5/2} r^{-1/2} < 0, \qquad \frac{\partial L^*}{\partial r} = \frac{p^2}{16}\Big(-\frac{1}{2}\Big) w^{-3/2} r^{-3/2} < 0.$$

This fits Cramer's rule: $f''_{LK} = \frac{1}{16} K^{-3/4} L^{-3/4} > 0$, so $\partial L^*/\partial r = -p f''_{LK}/|H| < 0$. But the conditional demand $y^2 (r/w)^{1/2}$ *rises* with r. The difference is output. Holding y fixed, pricier capital pushes the firm toward labor. Letting y adjust, pricier capital raises marginal cost, the firm shrinks, and it hires less of everything. Here the second effect wins.

(d) At $p = 8$, $w = r = 1$ we have $y^* = 2$.

- Left side, from (c): $\frac{64}{16} \cdot \big(-\frac{1}{2}\big) = -2$.
- Substitution, y fixed: $\frac{\partial}{\partial r}\big[y^2 r^{1/2} w^{-1/2}\big] = \frac{1}{2} y^2 r^{-1/2} w^{-1/2} = \frac{1}{2} \cdot 4 = 2$.
- Scale, first piece: $\frac{\partial}{\partial y}\big[y^2 (r/w)^{1/2}\big] = 2y(r/w)^{1/2} = 4$.
- Scale, second piece: $y^* = \frac{p}{4} w^{-1/2} r^{-1/2}$, so $\frac{\partial y^*}{\partial r} = \frac{p}{4} w^{-1/2}\big(-\frac{1}{2}\big) r^{-3/2} = 2 \cdot \big(-\frac{1}{2}\big) = -1$.

Right side: $2 + 4 \cdot (-1) = -2$. It matches. This is the firm's version of Slutsky: a substitution effect (positive here) plus a scale effect that works through output (negative and larger here).